

Gauge-Invariant Entanglement in a Minimal Fermionic Toy: The Coupling–Entropy Question Made Well-Posed — and Closed

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Abstract

A companion paper [1] showed that the question “is the Standard Model coupling hierarchy encoded in three-qubit vacuum entanglement?” is ill-posed: gauge transformations of the underlying fermionic modes are not local unitaries of the Jordan–Wigner qubit factorization, so single-qubit entropies are gauge-frame dependent. This paper makes the question well-posed and answers it. We replace subsystem entropies with entanglement relative to the gauge-invariant operator algebra — the commutant of the gauge action, in the algebra-relative framework of Barnum–Knill–Ortiz–Viola and Zanardi, which is the discrete cousin of the Casini–Huerta center decomposition for gauge fields. The commutants are computed numerically and identified exactly: gauging $U(3)$ yields the abelian algebra $\mathbb{C}^4$ of charge-sector projectors; $SU(3)$ yields $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$, with the M_2 block on $\text{span}\{\nu, e^+\}$; $U(1)$ yields $M_1 \oplus M_3 \oplus M_3 \oplus M_1$. Findings, all machine-verified with deterministic seeds: (i) the single-copy gauge-invariant data of any pure state under $U(3)$ is its electric-charge-sector distribution alone; (ii) for every pure state and all three algebras the quantum piece of the algebraic entropy vanishes identically — the toy’s gauge-invariant entanglement is 100% classical center term, the discrete analog of the edge-mode contribution in lattice gauge theory (“all edge, no bulk” — with the caveat, developed in Section 7.2, that the physical status of the edge term is itself contested [15]); (iii) the companion paper’s one surviving static “match” — the weighted-W coupling-matching curve — is pure gauge: every state on it has $S_{\mathcal{A}} = 0$ for all three algebras and is gauge-equivalent to a single Fock basis state, with $W = G(U)|001\rangle$ exhibited explicitly; (iv) the algebra-canonical rotor vacuum’s invariant data is exactly $(1/2, 3/14, 3/14, 1/14)$ with $p_1 = p_2$, i.e. two independent invariant parameters — fewer than three couplings; (v) an exhaustive search over 60 sector-assignment $\times$ feature-map combinations finds nothing within 0.05 of $r_{\text{SM}} = 1.8174$, the best unprincipled miss being $r = 2$ exactly; (vi) $SU(2)_L$ is not representable on the one-generation ideal at all — the mode gauge group is $U(3) \cong (SU(3)_c \times U(1)_{\text{em}})/\mathbb{Z}_3$ — so the three-coupling question was never posable in this toy. The well-posed question has a sharper negative answer than the ill-posed one it replaces. We state plainly what would have to exist for any version to survive: a two-ideal construction on which $SU(2)_L$ acts, and a principle linking sector statistics to couplings. Neither currently exists.

1 Introduction

The companion paper [1] asked whether the Standard Model gauge coupling hierarchy is encoded in the entanglement entropies of a three-qubit vacuum state, motivated by the convergence of the emergent-spacetime and division-algebraic programs through the Szangolies construction [5]. Its answer was a battery of no-go results, capped by a diagnosis: the question is *ill-posed*. Recognizing Furey’s $\text{Cl}(6)$ construction [2, 3, 4] as the three-qubit operator algebra under the Jordan–Wigner transformation [6] makes electric charge the Hamming-weight grading of the computational basis — and exposes the obstruction that gauge transformations act on the fermionic *modes*, not on the qubit tensor factors. A color discrete Fourier transform maps a separable single-quark state, with single-qubit entropies $(0, 0, 0)$, to the W state, with entropies $(0.918, 0.918, 0.918)$. The program’s central quantity was frame decoration.

The companion paper closed by identifying the construction of a gauge-invariant entanglement functional as the central open problem ([1], Secs. 9.3, 11.2). This paper completes that program for the minimal toy, and reports what the completed version says.

The functional is not new and we do not claim it as such: entanglement relative to a distinguished $*$ -subalgebra of observables is the “generalized entanglement” of Barnum, Knill, Ortiz, and Viola [8], the algebra-relative subsystem notion of Zanardi [7], and — in the form most relevant here — the algebraic definition that Casini, Huerta, and Rosabal introduced for gauge fields [9], whose classical center term Donnelly identified with lattice-gauge edge modes [10, 11]. Our contribution is to *execute* this definition exactly, end to end, in the one system where the companion paper’s question lives: the eight-dimensional one-generation ideal $(\nu, \bar{d}, u, e^+)$ in its Jordan–Wigner three-qubit presentation, under the three natural gauge choices $U(3)$, $SU(3)$, and $U(1)$. Everything is finite-dimensional, so there are no regularization choices to argue about; the commutants are computed numerically and certified, the entropies have closed forms, and every claim in this paper is asserted at runtime by the published code.

1.1 Contributions

1. **The gauge-invariant algebras, computed and identified exactly** (Section 4). Gauging $U(3)$ leaves only the abelian algebra $\mathbb{C}^4$ of charge-sector projectors; $SU(3)$ leaves $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ (dimension 6), the M_2 block living on $\text{span}\{\nu, e^+\}$ — the ν – e^+ coherence is gauge-invariant when only color is gauged; $U(1)$ leaves $M_1 \oplus M_3 \oplus M_3 \oplus M_1$ (dimension 20). Under the full $U(3)$, the single-copy gauge-invariant data of any pure state is its charge-sector distribution and nothing else.
2. **All edge, no bulk** (Section 5). For every pure state and all three algebras, the quantum piece of the algebraic entanglement entropy vanishes identically — a two-line consequence of the block structure, verified numerically to 1.8×10^{-15} . The toy’s entire gauge-invariant entanglement is the classical center term, the discrete analog of the edge-mode contribution in lattice gauge theory. As a corollary we derive $S_{U(1)} = S_{U(3)}$ for every pure state, despite the $U(1)$ -invariant algebra being five times larger.
3. **The headline: the companion paper’s surviving “match” is pure gauge** (Section 5.4). Every point of the weighted- W coupling-matching curve — the one static construction in [1] that reproduced $r_{\text{SM}} = 1.8174$ — has $S_{\mathcal{A}} = 0$ for all three invariant algebras and is gauge-equivalent to a single Fock basis state: one quark in a rotated color frame. We exhibit $U \in SU(3)$ with $G(U)|001\rangle = |W\rangle$ exactly. The static coupling-matching analysis of [1], Secs. 5–6, was matching frame artifacts.

4. **The well-posed question, answered** (Section 6). The gauge-invariant reformulation of the coupling-entropy question is stated and answered in the negative for four independent, machine-verified reasons: the invariant sectors are charge sectors, not gauge sectors; the algebra-canonical rotor vacuum carries exactly two independent invariant parameters, $p = (1/2, 3/14, 3/14, 1/14)$ with $p_1 = p_2$; exhaustive search over sector-to-coupling assignments misses r_{SM} (best unprincipled miss: $r = 2$ exactly, flagged as numerology bait); and $SU(2)_L$ admits no representation on the one-generation ideal — a machine-checked lemma (Appendix B.1), not only a structural argument — so a three-coupling question cannot even be posed.
5. **The bridge to the edge-mode literature, stated with its contested status on the surface** (Section 7.2). The center term computed throughout is the same object, in an 8-dimensional toy, as the classical flux-sector term of lattice gauge theory [9, 10, 11] and the edge-mode reading of Kabat’s contact term [12, 13, 14] — and we record honestly, following Casini–Huerta–Magán–Pontello [15], that the physical status of that term is itself contested in the continuum.

1.2 Scope

This paper is about an eight-dimensional toy. Nothing here bears on continuum gauge theory except by structural analogy, and the analogy’s load-bearing limit is stated where it is used. The no-go results are exact within the toy and claim nothing beyond it; their value, if any, is that they close the minimal case cleanly and locate exactly what a non-minimal case would need.

2 The Toy and Its Gauge Action

Status: construction inherited from [1], Sec. 9; machine-verified here to stated residuals.

The arena is $\mathbb{C}^8 = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$ with three Jordan–Wigner fermionic modes

$$f_1 = \sigma^- \otimes I \otimes I, \quad f_2 = Z \otimes \sigma^- \otimes I, \quad f_3 = Z \otimes Z \otimes \sigma^-,$$

satisfying the canonical anticommutation relations exactly (residual 0.0 in our run). The computational basis is the Fock basis, graded by Hamming weight, and the weight is three times the electric charge on Furey’s one-generation ideal [2, 3, 4]: ν at weight 0, the $\bar{d}$ color triplet at weight 1, the u triplet at weight 2, e^+ at weight 3, with $Q = N/3$ for N the number operator.

A mode rotation $U \in U(3)$, $f_b^\dagger \mapsto \sum_a U_{ab} f_a^\dagger$, induces the number-conserving unitary $G(U)$ on $\mathbb{C}^8$ acting blockwise on the weight sectors:

$$\text{weight 0: } 1, \quad \text{weight 1: } U, \quad \text{weight 2: } \Lambda^2 U, \quad \text{weight 3: } \det U,$$

with $G(e^{i\theta} \mathbf{1}) = e^{i\theta N}$ on the $U(1)$ subgroup. The construction was verified to machine precision: G unitary (residual 2.4×10^{-15}), homomorphism $G(U)G(V) = G(UV)$ (6.8×10^{-16}), and the mode-transformation property $G(U)f_b^\dagger G(U)^\dagger = \sum_a U_{ab} f_a^\dagger$ (1.9×10^{-15}).

Three gauge choices are considered, each physically motivated: the full **$U(3)$** (color and electromagnetism together — the largest group of number-conserving mode rotations); **$SU(3)$** alone (gauge color, leave charge ungauged); and **$U(1)$** alone (charge superselection without color gauging). The group $U(3) \cong (SU(3)_c \times U(1)_{\text{em}})/\mathbb{Z}_3$ is the toy’s entire mode gauge group; Section 6.5 explains why $SU(2)_L$ is not on this list and cannot be.

3 Entanglement Relative to a $*$ -Algebra

3.1 Definition

Let $\mathcal{A}$ be a unital $*$ -subalgebra of $M_d(\mathbb{C})$. By the Wedderburn–Artin theorem there is a unitary change of basis under which

$$\mathcal{H} \cong \bigoplus_k \mathbb{C}^{n_k} \otimes \mathbb{C}^{m_k}, \quad \mathcal{A} = \bigoplus_k M_{n_k}(\mathbb{C}) \otimes \mathbf{1}_{m_k},$$

with commutant $\mathcal{A}' = \bigoplus_k \mathbf{1}_{n_k} \otimes M_{m_k}$ and center $Z(\mathcal{A}) = \mathcal{A} \cap \mathcal{A}' \cong \mathbb{C}^{\#\text{blocks}}$, spanned by the minimal central projections P_k . The reduced state of $|\psi\rangle$ on $\mathcal{A}$ is the conditional expectation $E_{\mathcal{A}}(|\psi\rangle\langle\psi|)$ — equivalently, the unique state ω on $\mathcal{A}$ with $\omega(a) = \langle\psi|a|\psi\rangle$. Writing

$$p_k = \langle\psi|P_k|\psi\rangle, \quad \rho_k = \frac{1}{p_k} \text{Tr}_{m_k}(P_k|\psi\rangle\langle\psi|P_k),$$

the **algebraic entanglement entropy** is

$$S_{\mathcal{A}}(\psi) = \underbrace{H(\{p_k\})}_{\text{classical (center) piece}} + \underbrace{\sum_k p_k S(\rho_k)}_{\text{quantum piece}}, \quad (1)$$

in bits throughout. This is the generalized entanglement of [8], the algebra-relative notion of [7], and — crucially — the discrete cousin of the Casini–Huerta–Rosabal definition for gauge fields [9]: when the algebra of a region in a lattice gauge theory has a nontrivial center (boundary electric fluxes), the entropy acquires exactly the $H(\{p_k\})$ term — the Shannon entropy of the boundary-flux superselection sectors, nowadays read as the edge-mode contribution [10, 11]. The center term computed throughout this paper is the same object in an 8-dimensional toy.

Two consistency limits: if $\mathcal{A} = B(\mathcal{H}_A) \otimes \mathbf{1}_B$ is a factor, $S_{\mathcal{A}} = S(\rho_A)$ and the center piece is zero — the standard entanglement entropy is recovered; if $\mathcal{A}$ is abelian (all $n_k = 1$), $S_{\mathcal{A}} = H(\{p_k\})$ is purely classical.

Pitfall (machine-cross-checked on every call). $S_{\mathcal{A}}$ is *not* the von Neumann entropy of the $d \times d$ matrix $E_{\mathcal{A}}(\rho)$; the two differ by $\sum_k p_k \log_2 m_k$, because $E_{\mathcal{A}}(\rho)$ spreads each block uniformly over its multiplicity factor. The library computes $S_{\mathcal{A}}$ both ways and asserts agreement to 10^{-8} .

3.2 Why This Is the Right Gauge-Invariant Functional

If a compact group acts on $\mathcal{H}$ by unitaries $G(g)$, the gauge-invariant observables are exactly the commutant $\mathcal{A} = \{G(g)\}'$. Conjugation by any $G(g)$ fixes $\mathcal{A}$ pointwise, so $E_{\mathcal{A}}(G\rho G^\dagger) = E_{\mathcal{A}}(\rho)$: $S_{\mathcal{A}}$ **is gauge-invariant by construction**. The quantity that was frame-dependent in [1] — single-qubit entropy — is replaced by one that provably cannot be. (Verified numerically below to $\sim 10^{-15}$, alongside the $O(1)$ shifts of the single-qubit entropies under the same transformations.)

Honest scope note. $E_{\mathcal{A}}(\rho)$ captures every statistic obtainable by *single-copy measurements of gauge-invariant observables* — under a charge superselection rule this is all that is operationally accessible [16]. The polynomial invariant ring of the state under G is strictly larger: it contains multi-copy invariants (computed for the rotor vacuum in Section 7.4) that are constants of the G -orbit but not functions of $E_{\mathcal{A}}(\rho)$. The well-posed question of Section 6 is phrased in terms of the operational single-copy data; the multi-copy ring is flagged as the remaining unexplored invariant content, not silently absorbed.

4 The Gauge-Invariant Algebras

Status: computed (seed 20260609; direction_A/results.json).

Method. For each gauge choice, 50 Haar samples $G(U_s)$ were drawn and the commutant $\{X : [X, G(U_s)] = 0 \ \forall s\}$ computed as the SVD nullspace of the stacked maps $X \mapsto (\mathbf{1} \otimes G - G^\top \otimes \mathbf{1}) \text{vec}(X)$. Each basis was certified against 100 fresh samples (maximum commutator residual $< 1.2 \times 10^{-15}$), checked to be a unital $*$ -algebra (closure residual $< 1.3 \times 10^{-15}$), and its Wedderburn structure read off from the center and per-block ranks. Nothing below relies on paper-and-pencil representation theory; the structures are extracted from the numerics, and then agree with it. (Appendix B.2 closes the remaining numerical gap: for each gauge choice, exhibited exactly-commuting elements span a space whose dimension equals the computed commutant dimension, so the identifications below are exact, not approximate.)

gauged group	$\dim \mathcal{A}$	structure (block [weight support])	center	invariant data of a pure state
$U(3)$	4	$M_1[w0] \oplus M_1 \otimes \mathbf{1}_3[w1] \oplus M_1 \otimes \mathbf{1}_3[w2] \oplus M_1[w3]$ — abelian $\mathbb{C}^4$	$\mathbb{C}^4$	(p_0, p_1, p_2, p_3) : 3 real parameters
$SU(3)$	6	$M_2 \otimes \mathbf{1}_1[w0 \cup w3] \oplus M_1 \otimes \mathbf{1}_3[w1] \oplus M_1 \otimes \mathbf{1}_3[w2]$	$\mathbb{C}^3$	(p_1, p_2) + pure qubit in the M_2 block: 4 parameters
$U(1)$	20	$M_1[w0] \oplus M_3[w1] \oplus M_3[w2] \oplus M_1[w3]$	$\mathbb{C}^4$	3 probabilities + two projective pure states in $\mathbb{C}^3$: 11 parameters

Readings, in turn:

- **$U(3)$: the commutant is abelian.** The four weight sectors carry the inequivalent irreps $(1, U, \Lambda^2 U, \det U)$, each with multiplicity one, so the invariant algebra is the sector projectors and nothing more. *The only single-copy gauge-invariant data of any state is its charge-sector distribution (p_0, p_1, p_2, p_3) .* This is the precise, gauge-invariant version of [1]’s “index ambiguity” (Sec. 2.3 there): there is no gauge-covariant tensor factorization because there are no gauge-invariant subsystems at all — only charge sectors.
- **$SU(3)$: the ν - e^+ coherence survives.** Weights 0 and 3 are both $SU(3)$ -trivial, so the commutant contains a full M_2 block on $\text{span}\{|000\rangle, |111\rangle\} = \text{span}\{\nu, e^+\}$ (machine check: the M_2 central projection equals the projector onto indices $\{0, 7\}$; assertion threshold 10^{-9} , measured residual $\leq 3.3 \times 10^{-15}$). When only color is gauged, the GHZ-type ν - e^+ coherence is gauge-invariant data. Weights 1 and 2 (the **3** and **$\bar{3}$**) remain inequivalent and contribute $\mathbb{C} \oplus \mathbb{C}$.
- **$U(1)$: everything weight-preserving.** $G(e^{i\theta} \mathbf{1}) = e^{i\theta N}$, so the commutant is $M_1 \oplus M_3 \oplus M_3 \oplus M_1$ (dimension 20) — the richest of the three, retaining the full within-sector color states. Parameter-count sanity check: a pure state in $\mathbb{C}^8$ has 14 real parameters; gauging $U(1)$ erases exactly the 3 relative phases between the four sectors: $14 - 3 = 11$. ✓

So under $U(3)$ the invariant data has exactly 3 real parameters — naive room for one per coupling. Section 6 is about why that room cannot be used.

5 Gauge-Invariant Entanglement of the Named States

Status: computed; closed forms machine-checked to 10^{-12} .

5.1 The Quantum Piece Vanishes Identically

Proposition 1. *For every pure state $|\psi\rangle \in \mathbb{C}^8$ and each of the three algebras above, the quantum piece of Eq. (1) vanishes: $S_{\mathcal{A}}(\psi) = H(\{p_k\})$.*

Proof. Every Wedderburn block of all three algebras has $n_k = 1$ or $m_k = 1$. If $n_k = 1$, then ρ_k is a 1×1 state and $S(\rho_k) = 0$ trivially. If $m_k = 1$, the multiplicity factor is trivial, so $P_k|\psi\rangle$ is a vector in the block and $\rho_k = P_k|\psi\rangle\langle\psi|P_k/p_k$ is a rank-one projector: $S(\rho_k) = 0$. $\square$

(Numerical confirmation: maximum |quantum piece| = 1.8×10^{-15} over the named states and a 200-state subsample of the 10^4 -state Haar ensemble run through the full block machinery; the remaining ensemble states were evaluated with a sector formula that agrees with the full machinery to 2.0×10^{-15} on that subsample.)

In Casini–Huerta language: for pure states, this toy is *all edge-mode term, no bulk term*. Any claim that “vacuum entanglement encodes couplings” in this framework is therefore, gauge-invariantly, a claim about classical charge statistics. Section 7.2 develops this bridge; here we note the scope limit on the surface: Proposition 1 is a statement about *pure* states. For mixed states the $SU(3)$ and $U(1)$ algebras can acquire nonzero quantum pieces (the $U(3)$ entropy stays classical for all states because the algebra is abelian); and the first system in this program where a nonzero quantum piece could appear for pure states needs blocks with both $n_k, m_k > 1$ — e.g. regional subalgebras, or the two-ideal extension of Section 7.3.

Corollary 2 ($S_{U(1)} = S_{U(3)}$ for every pure state). *The $U(1)$ and $U(3)$ commutants have the same center — the four weight projectors. By Proposition 1, both entropies equal $H(p_0, p_1, p_2, p_3)$.* $\square$

This holds even though the $U(1)$ -invariant algebra is five times larger: gauging the extra $SU(3)$ changes *which states are distinguishable* (the $U(1)$ data distinguishes W from $|001\rangle$; the $U(3)$ data does not) but not the entropy. Entropy is a very lossy summary of $E_{\mathcal{A}}(\rho)$ — a deflation worth keeping in view whenever a single number is asked to carry structural weight.

5.2 The Table

All entropies in bits; sector probabilities exact rationals (machine-checked against the numerics to 10^{-10}). The rightmost column shows the gauge-variant single-qubit entropies for contrast.

state		(p_0, p_1, p_2, p_3)	$S_{U(3)}$	$S_{SU(3)}$	$S_{U(1)}$	single-qubit S (gauge-variant)
rotor	vacuum	$(1/2, 3/14, 3/14, 1/14)$ exact	1.724408	1.413800	1.724408	(0.5917, 0.5917, 0.5917)
W	color-uniform $\bar{d}$	$(0, 1, 0, 0)$	0	0	0	(0.9183, 0.9183, 0.9183)
GHZ	$(\nu + e^+)/\sqrt{2}$	$(1/2, 0, 0, 1/2)$	1	0	1	(1, 1, 1)
weighted-W matching state		$(0, 1, 0, 0)$	0	0	0	(0.9995, 0.8580, 0.7802)
([1], Fig. 1c curve)						
Haar ensemble	—		1.580 ± 0.239	1.398 ± 0.162	$= S_{U(3)}$	—
(10^4) , mean $\pm$ std						

Cross-paper note (weighted-W row). The representative matching state quoted here, $S = (0.9995, 0.8580, 0.7802)$, is the first stored point of the exp2 matching curve (weighted_W.examples[0] in exp2_results.json, curve grid index 0). The companion paper’s Fig. 1c quotes a different representative, $S = (0.978, 0.895, 0.849)$ — the separately stored curve

midpoint (`weighted_W.example`, grid index 21 of 43). These are different points on the same one-parameter $r_S = 1.8174$ matching curve, not a discrepancy: both satisfy $r_S = r_{\text{SM}}$ to solver precision, both are verified weight-1-supported, and both are pure gauge, with $S_{\mathcal{A}} = 0$ for all three invariant algebras (machine-verified in `direction_A/strengthen.py`).

Exact closed forms for the rotor vacuum (machine-checked to 10^{-12}):

$$S_{U(3)} = S_{U(1)} = \frac{1}{2} + \frac{1}{2} \log_2 14 - \frac{3}{7} \log_2 3 = 1.724407817863 \text{ bits},$$

$$S_{SU(3)} = \frac{4}{7} (\log_2 7 - 2) + \frac{3}{7} \log_2 \frac{14}{3} = 1.413799564606 \text{ bits},$$

and the $SU(3)$ -invariant qubit in the M_2 (ν/e^+) block is pure, with block probability $4/7$ and state $\rho = \frac{1}{8} \begin{pmatrix} 7 & i\sqrt{7} \\ -i\sqrt{7} & 1 \end{pmatrix}$ — the gauge-invariant ν - e^+ coherence.

Gauge invariance verified (the whole point). For every named state, every algebra, and 20 random elements of the corresponding gauge group: $\max |\Delta S_{\mathcal{A}}| < 3 \times 10^{-15}$ (assertion threshold 10^{-9}), while the single-qubit entropies of the *same* transformed states shift by up to 0.9226 bits. Bonus, also verified to $\sim 10^{-15}$: $S_{SU(3)}$ and $S_{U(1)}$ are invariant under the *full* $U(3)$ — $G(U)$ normalizes the weight decomposition, and the extra $U(1)$ phases are isospectral on the M_2 block.

5.3 State by State

The W state’s entanglement is pure gauge. We exhibit $U \in SU(3)$ (third column $(1, 1, 1)/\sqrt{3}$) with $G(U)|001\rangle = |W\rangle$ exactly (construction residual 0.0), and verify $E_{\mathcal{A}}(W) = E_{\mathcal{A}}(|001\rangle)$ for the $U(3)$ algebra (residual 2.6×10^{-16}). The paradox that opened the well-posedness diagnosis in [1], Sec. 9.3 — a separable quark state mapped to the “entangled” W by a color DFT — is here resolved rather than flagged: the algebraic entropy assigns **zero to both**, because a single quark in a color-rotated frame is still a single quark. The 0.918 bits the W state carries in the qubit factorization is entirely frame.

GHZ is one classical bit — or nothing. Under $U(3)$, which contains the electromagnetic $U(1)$, the GHZ state’s invariant content is the charge distribution $(1/2, 0, 0, 1/2)$: exactly one bit of classical charge uncertainty, no quantum piece. Under $SU(3)$ only, the ν - e^+ coherence sits inside the M_2 invariant block as a *pure* qubit, and $S_{SU(3)}(\text{GHZ}) = 0$. The “maximal tripartite entanglement” of GHZ is, gauge-invariantly, either one classical bit (charge superselection imposed) or no entropy at all (color-only gauging). Nothing about it is three-partite in any invariant sense. The toy cleanly separates color gauging from charge superselection here: the ν - e^+ coherence is physical data if and only if the abelian factor is not gauged.

The rotor vacuum — the unique ground state of the companion paper’s algebra-canonical Hamiltonian $H = i(L_u + 2R_u)$ ([1], Sec. 8) — is the only state in the list with nontrivial invariant data, and that data is two numbers (Section 6.3). Its invariant entropy, 1.724 bits, is comfortably *generic*: the Haar mean is 1.580 ± 0.239 . Gauge-invariantly, the canonical vacuum is not even unusually entangled.

5.4 The Headline: the Coupling-Matching Curve Collapses to Pure Gauge

The companion paper’s only surviving static “match” was the weighted-W family: a continuous curve of states $a|001\rangle + b|010\rangle + c|100\rangle$ reproducing the Standard Model log-inverse-coupling gap ratio $r_{\text{SM}} = 1.8174$ exactly under the two-parameter logarithmic map ([1], Eq. 1; Secs. 5.3, 6.2,

Fig. 1c). Every state on that curve is a weight-1 state. Its invariant data is therefore $p = (0, 1, 0, 0)$, identical to that of the Fock state $|001\rangle$, and

$$S_{\mathcal{A}} = 0 \text{ on the entire curve, for all three algebras}$$

(verified on all five stored curve points; $\max |S_{\mathcal{A}}| = 1.9 \times 10^{-15}$ — and exact by the support argument, since the conclusion depends only on the curve lying inside the weight-1 sector). The entire matching curve is one $U(3)$ gauge orbit point: gauge-equivalent to a single $\bar{d}$ quark in a rotated color frame. The gauge equivalence, too, holds for the whole curve by a one-line argument, not by sampling: $SU(3)$ acts transitively on the unit sphere of $\mathbb{C}^3$, so any unit weight-1 amplitude vector (a, b, c) is the third column of some $U \in SU(3)$, and the corresponding state is $G(U)|001\rangle$ exactly — the five stored points are numerical confirmation, not the proof. *The static coupling-matching analysis of [1], Secs. 5–6, was matching gauge artifacts.* The companion paper diagnosed its title question as ill-posed; the well-posed functional goes further and evaluates the would-be evidence at exactly zero.

5.5 Haar Ensemble and Analytic Cross-Check

Over 10^4 Haar-random pure states: $S_{U(3)}$ has mean 1.5798, std 0.2391, range $[0.4103, 1.9984]$; $S_{SU(3)}$ has mean 1.3981, std 0.1617, range $[0.4066, 1.5849]$. These match the analytic values: for Haar states the sector probabilities are Dirichlet-distributed with parameters the sector dimensions, $p \sim \text{Dir}(1, 3, 3, 1)$, giving $\mathbb{E}[S_{U(3)}] = 1.5767$ via digamma sums; merging the $SU(3)$ -trivial weights $\{0, 3\}$ gives $\text{Dir}(2, 3, 3)$ and $\mathbb{E}[S_{SU(3)}] = 1.3963$. Sampled-vs-analytic discrepancies $(0.003, 0.002)$ are consistent with $N = 10^4$ fluctuations.

6 The Well-Posed Question and Its Answer

6.1 The Reformulation

Well-posed question. Let $\mathcal{A}$ be the commutant of the gauge action on the one-generation ideal. Does there exist a *natural* map from the single-copy gauge-invariant data $E_{\mathcal{A}}(|\Omega\rangle\langle\Omega|)$ of an algebra-selected vacuum $|\Omega\rangle$ to the three Standard Model couplings $(\alpha_3, \alpha_2, \alpha_1)$ — “natural” meaning the assignment of invariant parameters to gauge factors is dictated by the algebra, not chosen by hand?

This is well-posed where the title question of [1] was not: $E_{\mathcal{A}}$ is gauge-invariant by construction (verified to 10^{-15}), and “how many independent parameters are available” is now a theorem rather than a frame choice. Under $U(3)$ the invariant data has exactly 3 real parameters — so there is naive room, one per coupling. The answer is nevertheless no, for four independent reasons, each machine-verified.

6.2 Reason 1: the Sectors Are Charge Sectors, Not Gauge Sectors

The center of the $U(3)$ -invariant algebra is spanned by the projectors onto eigenspaces of the *single operator* N (electric charge $Q = N/3$). The four labels $\{0, 1/3, 2/3, 1\}$ are values of one $U(1)$ quantum number. The $SU(3)$ factor contributes no additional center label — weights 1 and 2 carry the $\mathbf{3}$ and $\bar{\mathbf{3}}$, but that is representation *type*, not a continuous parameter — and there are four sectors for three would-be couplings. A map $(p_1, p_2, p_3) \rightarrow (\alpha_3, \alpha_2, \alpha_1)$ requires a bijection $\{\text{charge sectors}\} \rightarrow \{\text{gauge factors}\}$ that nothing in the algebra provides: the invariant data simply

does not factorize along gauge factors. This is the companion paper’s “index ambiguity” ([1], Sec. 2.3) promoted from labeling worry to structural fact.

6.3 Reason 2: the Algebra-Canonical Vacuum Has Only Two Parameters

The rotor vacuum’s invariant data is exactly

$$p = \left(\frac{1}{2}, \frac{3}{14}, \frac{3}{14}, \frac{1}{14}\right), \quad \text{with } p_1 = p_2 \text{ exactly,}$$

because the vacuum $(\sqrt{7}e_0 - iu)/\sqrt{14}$ has uniform $|\text{amplitude}|^2 = 1/14$ on all seven imaginary units, forcing p_k proportional to the sector dimension on weights 1–3 (machine check: $|p_1 - p_2| < 10^{-16}$; the rationals certified against the floats to 10^{-10}). The canonical vacuum therefore carries **two independent invariant parameters — fewer than three couplings**. The permutation degeneracy that killed every hierarchical reading in the frame-dependent analysis ([1], Secs. 6.2, 8) *survives gauge-invariantization* as an exact sector degeneracy. The obstruction was never the frame; it is the algebra’s symmetry. We stress that the parameter count is decisive jointly with the naturality requirement of Section 6.1 and the ansatz class inherited from [1] — two invariant parameters cannot fill three coupling slots under any algebra-dictated assignment within that class — not as bare dimension counting, which a sufficiently contrived map could always evade.

6.4 Reason 3: No Assignment Reproduces the Hierarchy Anyway

Even granting an unprincipled sector-to-coupling assignment, the numbers refuse. We tested every ordered assignment of three distinct charge sectors to the three couplings under three feature maps $(p_k, \log_2 p_k, -p_k \log_2 p_k)$ — $3 \times 24 = 72$ combinations, of which 60 yield a finite gap ratio (the $p_1 = p_2$ degeneracy renders 12 indeterminate) — against the inherited two-parameter logarithmic map ([1], Eq. 1), i.e. demanding $r = (x_a - x_b)/(x_b - x_c)$ equal $r_{\text{SM}} = 1.8174$ ([1], Eqs. 2–3; GUT-normalized α_1 , a convention this target inherits). Result: **none lands within the companion paper’s widened tolerance 0.05** (let alone the 0.01 of its Experiment 1); the best miss is

$$r = 2 \text{ exactly (feature } p, \text{ descending sectors } (0, 1, 3) \text{ or } (0, 2, 3)), \quad \text{distance } 0.1826.$$

We flag this near-miss explicitly so nobody rediscovers it as a “signal”: $r = 2$ versus 1.8174 is a 10% coincidence of small fractions — $(1/2 - 3/14)/(3/14 - 1/14) = 2$ — reachable only through assignments that, absurdly, put the *neutrino* sector and the *positron* sector on two different gauge couplings while skipping one of the two exactly degenerate quark sectors. Assignments using both quark sectors give $r \in \{0, -1, \pm\infty\}$, degenerate by $p_1 = p_2$. This is numerology bait, not structure. (The no-match conclusion is also normalization-independent: Appendix B.3 reruns the identical 60-assignment zoo with the abelian coupling in hypercharge normalization, target $r_Y = 1.0433$ instead of the GUT-normalized 1.8174, and again finds nothing within 0.05 — with a *worse* best miss, 0.2533.)

6.5 Reason 4: $SU(2)_L$ Is Not in the Toy at All

Status: machine-checked lemma (Appendix B.1) plus structural identification from Furey’s construction.

The mode-rotation (number-conserving Bogoliubov) gauge group of three Jordan–Wigner modes is exactly $U(3) \cong (SU(3)_c \times U(1)_{\text{em}})/\mathbb{Z}_3$ — there is no room for an independent $SU(2)$ acting on the color modes. This claim is no longer a representation-theoretic argument but a machine-checked lemma: Appendix B.1 verifies that the Hermitian generators preserving the mode span form a real

Lie algebra of dimension exactly $10 = \dim u(3) + 1$, each exponentiating to $e^{i\varphi}G(U)$ with $U \in U(3)$ (residuals $< 3 \times 10^{-15}$); that the centralizer of the color action within this Lie algebra is the two-dimensional *abelian* algebra $\text{span}\{\mathbf{1}, N\}$, so no $su(2)$ commuting with $SU(3)_c$ exists; and that enlarging to number-violating Bogoliubov generators only adds directions that break the Hamming-weight (electric-charge) grading. More physically, the ideal $(\nu, \bar{d}_i, u_i, e^+)$ contains no weak-isospin doublet pair: ν_L pairs with e_L^- (absent — the ideal contains e^+), u_L with d_L (absent — the ideal contains $\bar{d}$). Weak raising and lowering operators would map this ideal into a *different* ideal; in Furey’s full construction $SU(2)_L$ acts on the quaternionic factor of $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ — between ideals, never within one [3]. And the toy’s abelian charge is electromagnetic $Q = N/3$, not hypercharge: $SU(2)_L$ does not commute with Q , so any implementation would have to move states between Hamming-weight sectors, and the only weight-0/weight-3 pair, (ν, e^+) , is not a doublet.

Consequently *the three-coupling question cannot even be posed gauge-covariantly on a single generation ideal*: at most two gauge factors ($SU(3)_c$, $U(1)_{\text{em}}$) act, and their joint invariant data is the charge distribution. The companion paper asked whether three couplings are encoded in this system; the gauge-covariant accounting says the system never had three couplings’ worth of symmetry to begin with.

6.6 Verdict: More Constrained, Not Less

The ill-posed version of the question was *underconstrained*: a 14-real-parameter state space, a guaranteed-nonempty one-parameter matching manifold, statistically generic matches, W-class realizations ([1], Secs. 5.3, 6.1–6.2). The well-posed version is drastically *more* constrained:

- the invariant state space shrinks from 14 parameters to 3 ($U(3)$);
- the entire weighted-W matching mechanism is annihilated — $S_{\mathcal{A}} = 0$ on the whole curve;
- the algebra-selected vacuum’s data is fixed and degenerate: two parameters, $p_1 = p_2$;
- no natural sector $\rightarrow$ factor map exists, and the exhaustive unnatural ones miss r_{SM} ;
- one of the three couplings has no corresponding symmetry in the toy at all.

Answer to the re-posed question: no — and now provably, not just diagnostically. The gauge-invariant entanglement data of a single pure state in this toy is its electric-charge statistics. Charge statistics has no natural three-fold structure aligned with the three gauge factors; the canonical vacuum’s statistics are two-parameter degenerate; and $SU(2)_L$ is absent from the construction. The coupling hierarchy is not, and cannot be, encoded in the gauge-invariant entanglement of a single pure state of this toy.

7 Discussion

7.1 The No-Go Ladder

Across the two papers, the coupling-entropy question has now descended a complete ladder, each rung sharper than the last:

1. **Frame-dependent** ([1], Secs. 5–6): the static matching analysis succeeded too easily — matching was statistically generic, null controls matched equally well, and the one algebraically flavored success (the weighted-W curve) survived.

2. **Ill-posed** ([1], Sec. 9.3): the quantity being matched, single-qubit entropy of the Jordan–Wigner factorization, is not gauge-invariant. Diagnosis, not yet answer.
3. **Well-posed** (this paper, Secs. 3–4): replace subsystem entropy by entanglement relative to the gauge-invariant algebra. The question now has a definite, frame-independent content: three real parameters under $U(3)$.
4. **Still no** (this paper, Sec. 6): the well-posed question has a negative answer for four independent structural reasons — and it retroactively evaluates the rung-1 evidence at exactly zero (Sec. 5.4).

The pattern is worth stating because it is the opposite of the usual fear about no-go results: making the question well-posed did not rescue a positive answer that ill-posedness had been obscuring; it converted a diagnostic “the question is meaningless” into a computed “the question is meaningful and the answer is no.”

7.2 All Edge, No Bulk: the Bridge to the Continuum — With Its Dispute Included

For pure states in this toy, the algebraic entanglement entropy is 100% center term (Proposition 1) — the discrete analog of the statement that a lattice-gauge entanglement entropy is carried by the classical boundary-flux (edge-mode) distribution. The lineage of that object in the continuum is well documented: Kabat’s contact term in the conical entropy of gauge fields [12]; the Donnelly–Wall edge-mode program, which interpreted it as the Shannon entropy of normal-flux sectors [13, 14]; Donnelly’s lattice form, $S = H(\{p_R\}) + \sum_R p_R(\cdots)$, of which Eq. (1) is the abstract finite-dimensional version [10, 11]; and the Casini–Huerta–Rosabal algebra-with-center formulation that our definition discretizes directly [9]. The toy realizes this structure in its most extreme form: all edge, no bulk. Any future claim that vacuum entanglement encodes couplings in this framework is therefore, gauge-invariantly, a claim about precisely the object that literature studies as edge modes.

Honesty requires importing the dispute along with the structure. Casini, Huerta, Magán, and Pontello [15] argue that in the continuum the center/edge term is an artifact of the algebra (equivalently regularization) choice: local continuum algebras need not have centers, the classical term drops out of mutual information and relative entropy, and related work shows it is not distillable [17, 18]. On that view, the quantity this paper computes wholesale — $H(\{p_k\})$ — is exactly the piece of “entanglement entropy” whose physical status is contested. Three things can be said in the toy’s defense, none of them a refutation. First, in the toy the algebra is not a regularization choice: given the gauge group, the commutant is canonical — the ambiguity that worries [15] reappears here only as the (physically meaningful) choice of *which group to gauge*, and we computed all three. Second, under a charge superselection rule the center term has a clean single-copy operational reading [16]: it is the Shannon entropy of the charge measurement that superselected observers can actually perform. Third, and cutting the other way: if one adopts the strict Casini–Huerta–Magán–Pontello standard — only mutual-information-like quantities are physical — then the toy’s gauge-invariant entanglement is not merely classical but arguably *empty*, and the no-go of Section 6 becomes if anything stronger. The dispute does not threaten our negative conclusion; it threatens only the consolation prize.

One further contrast pre-empts a natural question and frames the sequel. Donnelly’s two-dimensional Yang–Mills results [10, 11] show that a gauge coupling *is* exactly recoverable from the flux-sector (center) distribution — the very object computed throughout this paper. But

that recovery is *reflection*, not determination: there, dynamics supplies a known one-parameter exponential family $p_R(g^2 A)$ of flux-sector probabilities, and a known family is statistically invertible given the theory — the coupling can be read off the sector statistics because the theory says how it got in. The no-go of Section 6 concerns *determination*: an algebra-dictated map from the invariant data of a single kinematic state to three coupling constants, with no dynamics supplying a family and one of the three gauge groups not acting on the construction at all. The two notions are complementary, not in tension — sector statistics can perfectly well reflect a coupling that dynamics put there, even though no kinematic invariant data determines couplings here. A companion paper develops the positive (reflection) case exactly, in two-dimensional Yang–Mills, where the coupling is not only reflected but efficiently estimable from the flux-sector statistics [20].

7.3 What Two Ideals Would Change

The single most direct continuation is the two-ideal extension: build the 16-dimensional two-ideal space (or the full $\text{Cl}(6)$ Fock space with Furey’s left/right ideal pair), implement $SU(2)_L$ as the ideal-mixing quaternionic action, and recompute the commutant of the full $S(U(3) \times U(2))$ -type action with the same machinery. That construction would be the first in this program on which a three-coupling question is even posable gauge-covariantly. The questions it must answer are exactly the ones the one-ideal no-go quantifies: does the center acquire a natural *threefold* structure aligned with gauge factors, or does the charge-sector obstruction of Section 6.2 persist? And do blocks with both $n_k, m_k > 1$ appear, so that a nonzero quantum piece — genuine gauge-invariant quantum entanglement, “bulk” — exists for pure states at all? We make no prediction beyond noting that every symmetry-based degeneracy found so far has survived each previous upgrade of the formalism.

7.4 Beyond Single Copies: the Remaining Invariant Content

The no-go of Section 6 governs single-copy gauge-invariant data — everything measurable under the superselection rule. The full polynomial invariant ring of a state under $G(U(3))$ is strictly larger, and for the rotor vacuum its low-degree elements are nonzero and exactly computable: with v_1, v_2 the weight-1 and weight-2 component vectors,

$$|v_1 \wedge v_2| = \frac{1}{14} \text{ exactly,} \quad |\bar{v}_3 \cdot (v_1 \wedge v_2)| = 14^{-3/2} \approx 0.019090, \quad |\bar{v}_0 v_3 \overline{(v_1 \wedge v_2)}| = \frac{\sqrt{7}}{196} \approx 0.013499,$$

all verified gauge-invariant to 5.6×10^{-17} under random $G(U)$. These are not functions of $(p_0, \dots, p_3)$: they are constants of the gauge orbit invisible to E_A . The single-copy no-go therefore does not automatically extend to the full invariant ring — though it does extend to anything operationally accessible under superselection, and we know of no principle that would route three coupling constants through multi-copy invariants while bypassing every single-copy observable. Characterizing the full $U(3)$ invariant ring on $\mathbb{C}^8$ (generator count via Jacobian rank at generic points) is concrete, undone work.

7.5 Limitations

The obvious one bears repeating: this is an 8-dimensional toy with no dynamics, no scale, and (Sec. 6.5) at most two gauge factors. The no-go results are exact but local to the construction; they constrain the minimal three-qubit literalism of the companion paper’s program, not gauge theory. The flow version of the question — whether any RG-consistent trajectory on the sector simplex, with the rotor vacuum’s $(1/2, 3/14, 3/14, 1/14)$ as boundary data, can reproduce one-loop

running — remains open, though Section 6.2 makes the expected outcome plain: the simplex has no natural three-coupling structure either. Finally, the $r_{\text{SM}} = 1.8174$ target inherits the GUT normalization $\alpha_1 = (5/3)\alpha_Y$ from [1]; the *target* of the exhaustive search of Section 6.4 is in that sense convention-dependent, but the outcome was checked directly and is not: with the abelian coupling in hypercharge normalization, $\alpha_Y^{-1} = (5/3)\alpha_1^{-1} = 98.33$, the target becomes $r_Y = 1.0433$, and the identical 60-assignment zoo again finds no match within 0.05 — best miss 0.2533, farther than the GUT-normalized 0.1826 (Appendix B.3). And the structural reasons (Secs. 6.2, 6.3, 6.5) never depended on the convention — $p_1 = p_2$ does not care how the abelian coupling is normalized.

8 Conclusion

The companion paper ended by declaring the coupling-entropy question ill-posed until a gauge-invariant entanglement functional was specified, and named that specification the program’s central open problem. This paper specified the functional — the standard algebraic one, executed exactly — and the completed program closes the question rather than reopening it. The well-posed question has a sharper negative answer than the ill-posed one: the gauge-invariant entanglement of any pure state of the minimal fermionic toy is its electric-charge statistics, entirely classical (“all edge, no bulk”); the algebra-canonical vacuum carries two independent invariant parameters, not three; the one positive-seeming static result of the companion paper is revealed to be a single quark in a rotated color frame, with invariant entropy exactly zero; and the weak gauge factor was never representable on the construction in the first place.

For any version of the coupling-entropy correspondence within this division-algebraic program to survive, two things would need to exist, neither of which currently does. First, a richer arena: a two-ideal (or larger) construction on which $SU(2)_L$ acts, so that a three-coupling question is gauge-covariantly posable at all — and on which one can then check whether the invariant center acquires a natural threefold structure aligned with gauge factors, or whether the charge-sector obstruction found here persists. Second, a principle: an algebra-dictated, not hand-chosen, map from gauge-invariant sector statistics to coupling constants, presumably with scale dependence built in. Absent both, the conclusion of the minimal case stands as computed: there is nothing in the gauge-invariant entanglement of this toy for the Standard Model couplings to be encoded in. We offer the closed case, the machinery (which applies unchanged to any finite-dimensional gauge action), and the precise specification of what a successor construction must provide.

A Reproducibility

Code. All computations in this paper are produced by three files in the repository’s `direction_A/` directory:

- `alg_entanglement.py` — library: numerical commutant (SVD nullspace of the stacked commutator maps), Wedderburn structure identification (center, minimal central projections, per-block ranks), trace-preserving conditional expectation, algebraic entropy with the built-in cross-check $S_{\mathcal{A}} = S(E_{\mathcal{A}}(\rho)) - \sum_k p_k \log_2 m_k$ asserted to 10^{-8} on every call, Jordan–Wigner modes, the $G(U)$ construction, and Haar samplers.
- `run_direction_A.py` — every computation reported in Sections 4–6, in order; fixed seed 20260609; every claim asserted at runtime, so the script fails if any verification breaks; writes `results.json`.

- **strengthen.py** — the machine-checked lemmas of Appendix B (the mode-gauge-group lemma with its Bogoliubov, particle-hole, and antilinear variants; the $SU(3)$ -centralizer computation; the hypercharge-normalization rerun of the assignment zoo; and the cross-paper weighted-W point reconciliation of Section 5.2); fixed seed 20260609; writes **strengthen.results.json**.

Run: `python run_direction_A.py` (≈ 6 s; NumPy + SciPy only), then `python strengthen.py` for Appendix B. The scripts read `../experiments/results/exp2.results.json` (published with [1]) for the (five) stored weighted-W matching-curve amplitudes tested in Section 5.4.

Sampling protocol. Commutants: 50 Haar generator samples per gauge group, certified against 100 fresh samples (max commutator residual $< 1.2 \times 10^{-15}$) and checked for unital $*$ -closure ($< 1.3 \times 10^{-15}$). Gauge invariance: 20 random gauge elements per state per group (assertion threshold 10^{-9} ; observed $< 3 \times 10^{-15}$). Haar ensemble: 10^4 states, with a 200-state subsample run through the full block machinery (fast-formula agreement 2.0×10^{-15} ; max |quantum piece| 1.8×10^{-15}) and analytic Dirichlet cross-checks of the means asserted to 0.01.

Exact rationals. Sector probabilities reported as fractions ($1/2$, $3/14$, $3/14$, $1/14$, etc.) are produced by `Fraction(x).limit_denominator( $10^4$ )` and asserted to agree with the floating-point values to 10^{-10} ; the closed-form entropies are asserted against the numerics to 10^{-12} ; $|v_1 \wedge v_2| = 1/14$ is asserted to 10^{-12} . The statement $p_1 = p_2$ is exact by construction (uniform |amplitude| $^2 = 1/14$ on the seven imaginary units forces $p_k \propto$ sector dimension on weights 1–3) and machine-checked to $< 10^{-16}$.

Construction residuals (this run): CAR relations 0.0; $G(U)$ unitarity 2.4×10^{-15} ; homomorphism 6.8×10^{-16} ; mode-transformation property 1.9×10^{-15} ; $W = G(U)|001\rangle$ construction 0.0, with E_A reduction agreement 2.6×10^{-16} ; M_2 -block support identification asserted at threshold 10^{-9} — an assertion threshold, not a measured residual (measured: $\leq 3.3 \times 10^{-15}$).

B Machine-Checked Lemmas

Three statements used in the main text are recorded here as machine-checked lemmas rather than numerical observations or structural arguments. Code: `direction_A/strengthen.py` (seed 20260609; NumPy + SciPy only); raw outputs in `direction_A/strengthen.results.json`; every claim below is asserted at runtime, so the script fails if any verification breaks.

B.1 The Mode-Preserving Unitary Group Is $\{e^{i\varphi}G(U) : U \in U(3)\}$

Lemma B.1. *Let $F = \text{span}_{\mathbb{C}}\{f_1, f_2, f_3\} \subset M_8(\mathbb{C})$ be the span of the three Jordan–Wigner modes. Then:*

(i) *(Lie algebra.) $L := \{h = h^\dagger : [h, f_k] \in F, k = 1, 2, 3\}$ has $\dim_{\mathbb{R}} L = 10 = \dim u(3) + 1$, and $L = d\Gamma(u(3)) \oplus \mathbb{R} \cdot \mathbf{1}$, where $d\Gamma(a) = \sum_{ij} a_{ij} f_i^\dagger f_j$. Every element of L commutes with the number operator N .*

(ii) *(Exponentiation.) For every $h \in L$, $W = e^{ih}$ satisfies $W f_k W^\dagger = \sum_j M_{kj} f_j$ with $M \in U(3)$, and $W = e^{i\varphi}G(M^\dagger)$ exactly.*

(iii) *(Converse — group level, all components.) Any unitary W with $W F W^\dagger \subseteq F$ is of the form $e^{i\varphi}G(U)$: the canonical anticommutation relations force the induced coefficient matrix M to be unitary, and $W G(M^\dagger)^\dagger$ commutes with every f_k and $f_k^\dagger$, which generate all of $M_8(\mathbb{C})$, so it is scalar by Schur’s lemma. No extra connected or disconnected components exist.*

(iv) *(The $SU(2)_L$ obstruction proper.) The centralizer of the color action $\{G(V) : V \in SU(3)\}$ inside L is $\text{span}_{\mathbb{R}}\{\mathbf{1}, N\}$: two-dimensional and abelian. An $SU(2)_L$ gauge factor would have to*

commute with $SU(3)_c$ and act on the modes; $su(2)$ needs a three-dimensional non-abelian algebra, and there is none.

Machine checks. The constraint $[h, f_k] \in F$ is linear in the 64 real parameters of a Hermitian 8×8 matrix; its solution space was computed as an SVD nullspace. The dimension count is unambiguous: largest “zero” singular value 1.1×10^{-15} versus smallest nonzero singular value 1.0. Constraint residual of the solution basis 9.3×10^{-16} ; $\|[h, N]\| \leq 1.1 \times 10^{-15}$ for every solution; the identification $L = d\Gamma(u(3)) \oplus \mathbb{R} \cdot \mathbf{1}$ is an exact rank check ($10 = 10$). Exponentiation was verified for all 10 basis elements and 20 random real combinations: $\max \|W f_k W^\dagger - \sum_j M_{kj} f_j\| = 2.8 \times 10^{-15}$, $\max \|M M^\dagger - \mathbf{1}\| = 4.7 \times 10^{-16}$, $\max \|W - e^{i\varphi} G(M^\dagger)\| = 2.5 \times 10^{-15}$ (all residuals $< 3 \times 10^{-15}$). The algebra generated by $\{f_k, f_k^\dagger\}$ has dimension $64 = \dim M_8(\mathbb{C})$, validating the Schur step. The $SU(3)$ -centralizer in L has dimension exactly 2, equals $\text{span}\{\mathbf{1}, N\}$ by exact rank check, is abelian (basis commutator $< 10^{-12}$), and was certified on 20 fresh Haar $SU(3)$ samples (residual 1.1×10^{-15}). This upgrades the $SU(2)_L$ -unrepresentability discussion of Section 6.5 from a representation-theoretic argument to a machine-checked lemma.

Bogoliubov, particle-hole, and antilinear variants (also machine-checked). Allowing number-violating maps $f_i \mapsto \sum_j (U_{ij} f_j + V_{ij} f_j^\dagger)$ — constraint $[h, f_k] \in F \oplus F^\dagger$ — enlarges the generator space to $\dim_{\mathbb{R}} 16 = \dim so(6) + 1$, identified exactly as $d\Gamma(u(3)) \oplus \mathbb{R} \cdot \mathbf{1}$ plus the six pair terms $f_i^\dagger f_j^\dagger + \text{h.c.}$ and $i(f_i^\dagger f_j^\dagger - \text{h.c.})$. But every added direction violates the Hamming-weight (electric-charge) grading: the number-conserving part collapses back to L exactly (dimension 10, identical span); the pair generator $f_1^\dagger f_2^\dagger + \text{h.c.}$ has $\|[h, N]\| = 4.0$, its commutator with f_1 has a component of norm 2.0 entirely outside F , and its exponential has off-weight-block norm 0.84 (it moves states across charge sectors). A particle-hole unitary exists — $X \otimes X \otimes X$ implements $f_k \mapsto (f_1^\dagger, -f_2^\dagger, f_3^\dagger)$ exactly (residual 0) — but it inverts the charge grading ($N \mapsto 3 - N$) and swaps $F \leftrightarrow F^\dagger$, so it lies outside the span-preserving group; $SU(2)$, being connected, cannot live in a disconnected coset. Antilinear maps add nothing new: all f_k are real matrices, so every antiunitary preserving F is entrywise conjugation K composed with a span-preserving unitary, and $KG(U)K = G(\bar{U})$ (residual 0).

B.2 The Commutant Identifications Are Exact (Sandwich Argument)

Lemma B.2. *For each gauge choice $\mathcal{G} \in \{U(3), SU(3), U(1)\}$, the commutant $\{G(g) : g \in \mathcal{G}\}'$ is exactly the algebra identified in Section 4: the abelian $\mathbb{C}^4$ of weight projectors (dimension 4), $M_2 \oplus \mathbb{C} \oplus \mathbb{C}$ with the M_2 block on $\text{span}\{|000\rangle, |111\rangle\}$ (dimension 6), and $M_1 \oplus M_3 \oplus M_3 \oplus M_1$ of all weight-preserving operators (dimension 20), respectively.*

Proof (sandwich). For each gauge choice, exhibit elements that commute with every $G(g)$ exactly, by construction: the four weight projectors ($U(3)$); the weight-1 and weight-2 projectors together with the four matrix units on $\text{span}\{|000\rangle, |111\rangle\}$, both weights being $SU(3)$ -trivial ($SU(3)$); and all weight-preserving matrix units ($U(1)$). These span spaces of dimensions 4, 6, and 20, giving lower bounds on the commutant dimensions. The numerically computed commutant — the SVD nullspace of the stacked commutator maps over the Haar samples — contains the true commutant, since commuting with the whole group implies commuting with the samples; its dimension is therefore an upper bound. The computed dimensions are 4, 6, and 20: the bounds coincide, forcing equality. The commutant is exactly the span of the exhibited elements, and the numerical identification is exact rather than approximate. $\square$

For $U(3)$ the identification also follows from Schur’s lemma alone: the four weight sectors carry the inequivalent irreps $1, U, \Lambda^2 U, \det U$, each with multiplicity one, so the commutant is the span of the four sector projectors.

B.3 The Zoo Miss Is Normalization-Independent

Lemma B.3 (computational). *With the abelian coupling in hypercharge normalization, $\alpha_Y^{-1} = (5/3)\alpha_1^{-1} = 98.33$, the target gap ratio becomes $r_Y = 1.0433$ in place of the GUT-normalized $r_{\text{SM}} = 1.8174$ — and the identical 60-assignment sector-to-coupling zoo of Section 6.4, run on the rotor vacuum’s exact invariant data $p = (1/2, 3/14, 3/14, 1/14)$ against r_Y with the same 0.05 window, again finds nothing: no assignment within 0.05, none within 0.01. The conclusion of Section 6.4 is normalization-independent.*

Machine checks. Inputs as in [1], from [19]: $\alpha_3 = 0.1179$, $\alpha_2 = 0.03374$, $\alpha_1 = 0.01695$ ($\alpha_1^{-1} = 58.9971$); $\alpha_Y^{-1} = (5/3) \times 58.9971 = 98.3284$, giving $r_Y = 1.043286$ (rounded-table variant 98.3333, $r_Y = 1.043508$). The reimplemented zoo reproduces the published GUT-target best miss (0.18256) as a cross-check. Against r_Y , the best miss is 0.2533, at $r = \log_2 3 / \log_2(7/3) = 1.296607$ (feature $\log_2 p_k$, sectors $(e^+, \bar{d}, \nu)$ and its exact mirror (e^+, u, ν) , degenerate by $p_1 = p_2$); rounded-target variant 0.2531. The α_Y miss is *worse* than the GUT miss (0.1826). Assignments using both quark sectors still give $r \in \{0, -1, \pm\infty\}$: the $p_1 = p_2$ degeneracy and the four-sectors-versus-three-couplings mismatch are untouched by the choice of abelian normalization, as expected.

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